

T04 INDIVIDUAL THERMAL CONTROL | O (MAX : 3 PT)

Intent : Maximize and personalize thermal comfort among all individuals.

Summary : This WELL feature requires projects to improve thermal comfort of people in the space through the provision of personal thermal comfort devices and flexible dress codes that support individual thermal preferences.

Issue : Many thermal comfort standards (such as ASHRAE 55) aim to provide a thermal environment, where at minimum 80% of people will perceive the thermal environment as acceptable.¹ Due to the individualized nature of thermal comfort, not everyone will be equally comfortable under the same environmental conditions. Besides the six primary factors of thermal comfort, many secondary factors, such as age, sex, health condition, personal thermal adaptation and thermal history, including climatological origin, are considered as major secondary factors.²⁻⁵ Factors, such as temperament, preferences, social and cultural norms and seasonal variation also play an important role in determining individual thermal comfort.^{6,7}

Solutions : Adopting holistic approaches through the provision of individual thermal control devices results in improved individual thermal comfort and in the expansion of the thermal comfort acceptability limits.^{6,8} Individual thermal control also allows for a broader range of recommended indoor air temperatures, which is linked to energy savings potential.⁹⁻¹¹ In larger shared spaces, use of personalized equipment is a good strategy to give people the ability to better control their sensation and comfort.¹²⁻¹⁴ For example, the use of a chair equipped with fans allows the room air temperature to increase without compromising thermal comfort.¹⁵⁻¹⁷ In addition, flexible dress codes are important, because clothing insulation is also one of the primary forms of intervention for addressing suboptimal thermal conditions.¹⁸

PART 1 PROVIDE PERSONAL COOLING OPTIONS (MAX : 1 PT)

For All Spaces except Dwelling Units:

The project provides all regular occupants with the ability to cool their individual environment through at least one of the following:¹

- a. A user-adjustable thermostat, which controls the environment for no more than one person and is connected to the building's mechanical cooling system or a more localized air conditioning unit.
- b. A desk, standing, pedestal or ceiling fan that does not increase air speed for other occupants.
- c. Chair with mechanical cooling system.
- d. Any other solution capable of affecting a PMV change of -0.5 within 15 minutes from activation, without changing the PMV for other occupants.

PART 2 PROVIDE PERSONAL HEATING OPTIONS (MAX : 1 PT)

For All Spaces except Commercial Kitchen Spaces & Dwelling Units:

The project provides all regular occupants with the ability to warm their individual environment through at least one of the following:¹

- a. A user-adjustable thermostat, which controls the environment for no more than one person and is connected to the building's mechanical heating system.
- b. Electric parabolic space heater.
- c. Electric heated chair or footwarmers.
- d. Personal or shared blankets. Shared blankets are washed or disinfected at least weekly after use.
- e. Any other solution capable of affecting a PMV change of +0.5 within 15 minutes from activation, without changing PMV for other occupants.

PART 3 ALLOW FLEXIBLE DRESS CODE (MAX : 1 PT)

For All Spaces except Commercial Kitchen Spaces & Dwelling Units:

The following requirement is met:

- a. A flexible dress code policy is in place that allows regular occupants to dress for individual thermal preferences.¹⁹

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